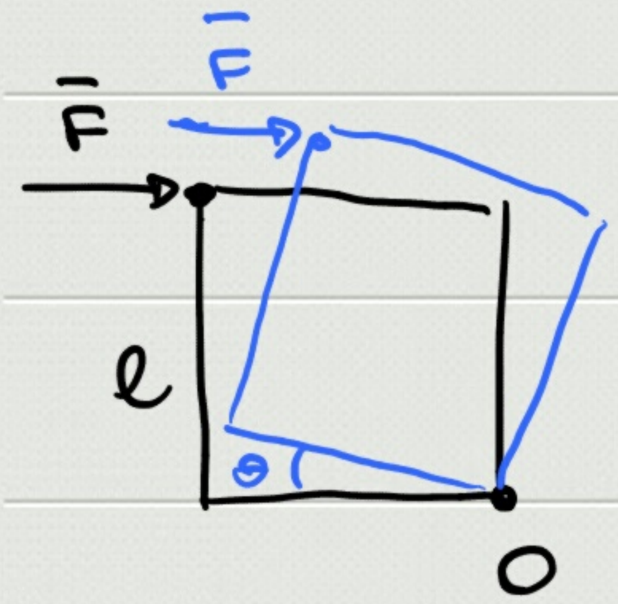


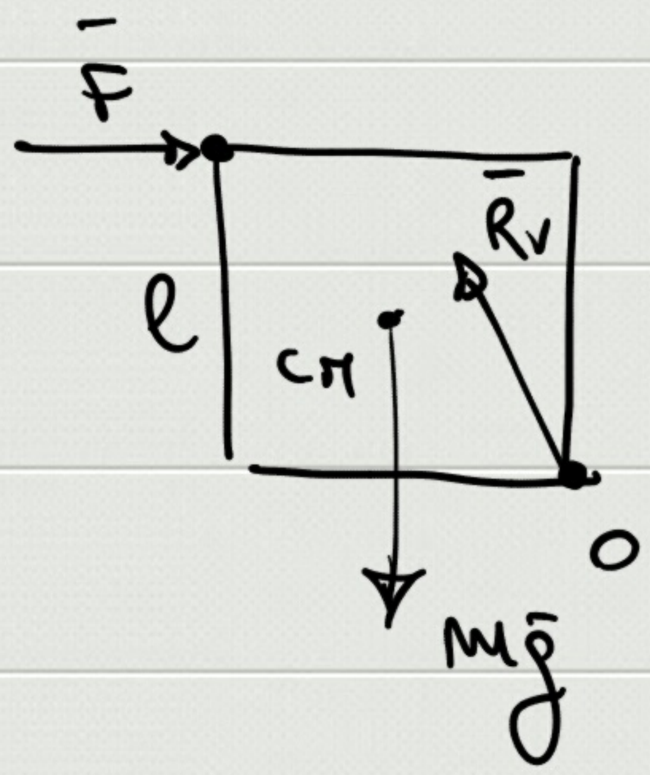
$\ell = 1.2\ m$

$m = 4 \cdot 10^3\ kg$

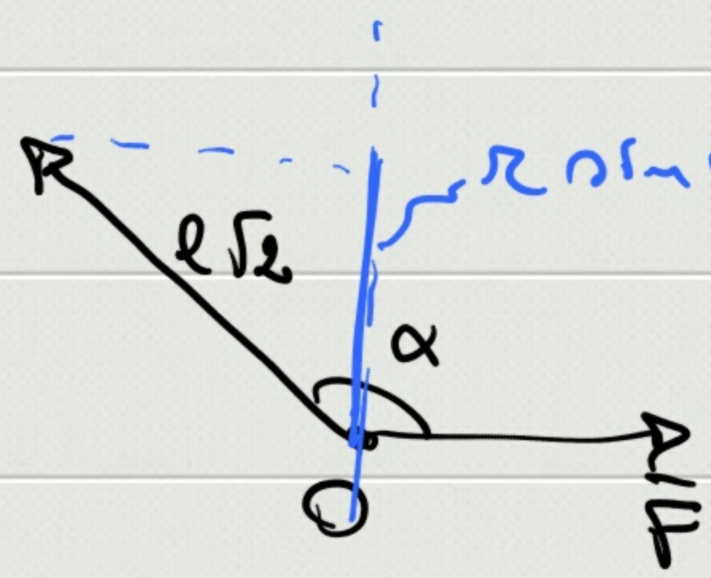
$F = 2 \cdot 10^4\ N$

$\alpha(t=0) = ?$

$$\boxed{\bar{M}_O^E = I_z \bar{\alpha}} \Rightarrow \alpha(t_0) = \frac{M_O^E(t_0)}{I_z}$$



$$\bar{M}_O = \bar{r} \times \bar{F} = r F \sin\alpha\ \bar{U}_M$$



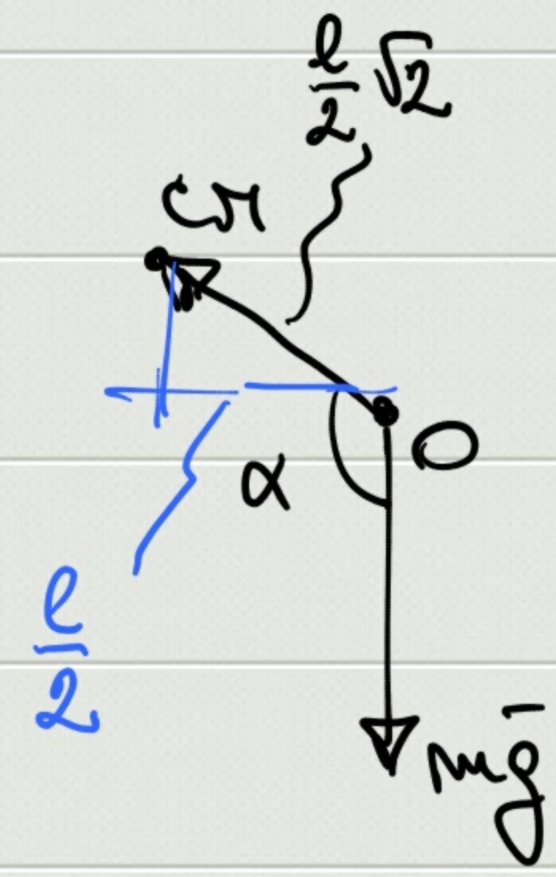
$r \sin\alpha = \ell\sqrt{2}\,\frac{\sqrt{2}}{2} = \ell$

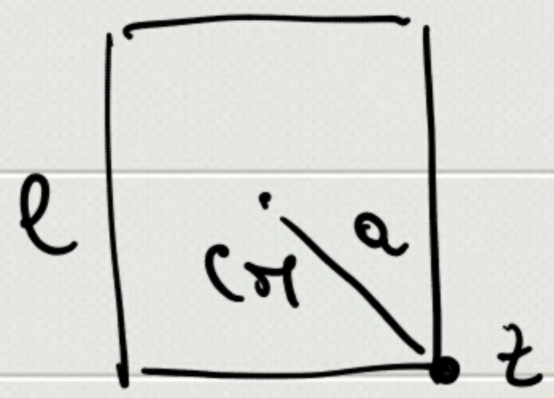
$$M_O = \cancel{\ell F \sqrt{2}}$$

$$M_O = \ell F - \frac{\ell}{2} mg \quad *$$

$$M_O = \cancel{\ell F \sqrt{2} - \frac{\ell}{2} mg \sqrt{2}}$$

$$M_O = \ell F + \frac{\ell}{2} mg$$





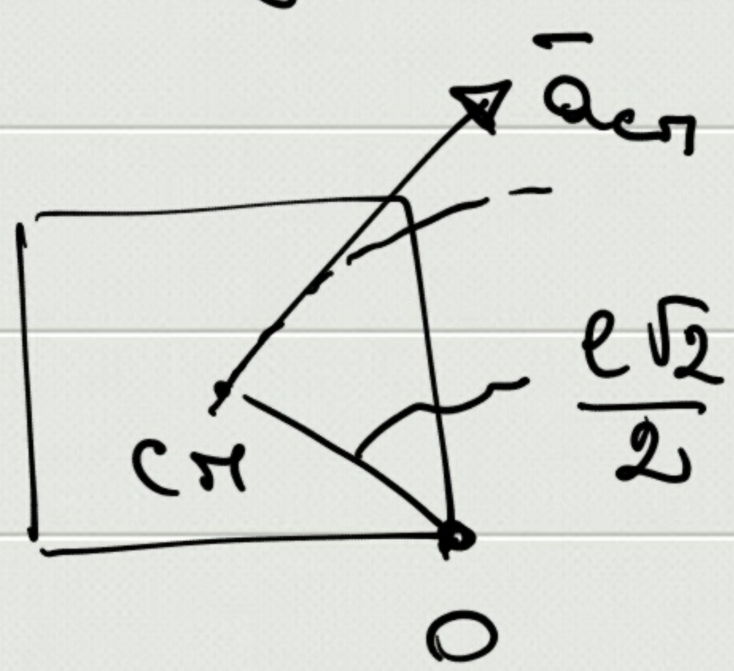
$$I_z = I_{z'} + m a^2$$

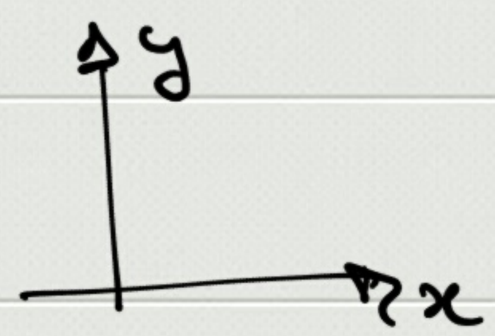
$$= \frac{1}{6} m l^2 + m \frac{l^2}{2} = \frac{2}{3} m l^2$$

$$\alpha(t_0) = \frac{M_0}{I_z} = \frac{6F - 3mg}{4ml} = 0{,}125 \ \text{rad/s}^2$$

$R_v = ?$

$$\bar{R}^e = m\bar{g} + \bar{F} + \bar{R}_v = m\bar{a}_{CM} \qquad a_{CM} = \alpha \frac{l\sqrt{2}}{2}$$





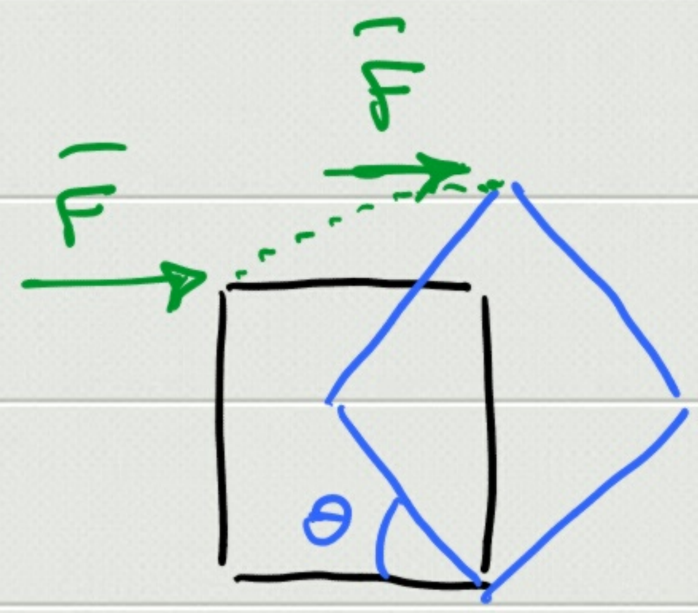
$$x: \ F + R_{v_x} = m a_{CM,x} = m\alpha \frac{l\cancel{\sqrt{2}}}{\cancel{2}} \frac{\cancel{\sqrt{2}}}{2} = m\alpha \frac{l}{2}$$

$$y: \ -mg + R_{v_y} = m a_{CM,y} = m\alpha \frac{l\sqrt{2}}{2} \frac{\sqrt{2}}{2} = m\alpha \frac{l}{2}$$

$$R_{rx} = m\alpha \frac{\ell}{2} - F = 1{,}97 \cdot 10^4 \text{ N}$$

$$R_{ry} = m\alpha \frac{\ell}{2} + mg = 3{,}95 \cdot 10^4 \text{ N}$$

$W_F(\theta = 45°) = ?$



$$W_F = F\,\ell\sqrt{2}\,\frac{\pi}{4}$$

$$W_F = F\,\ell\,\frac{\sqrt{2}}{2}$$

$$W_F = F\,\ell \quad *$$

$$W_F = 0$$

$$\bar{F} = F\,u_x$$

$$W = \int \bar{F}\,d\bar{s} \overset{!}{=} \int_0^{\ell} F\,dx$$

$$= F\ell$$

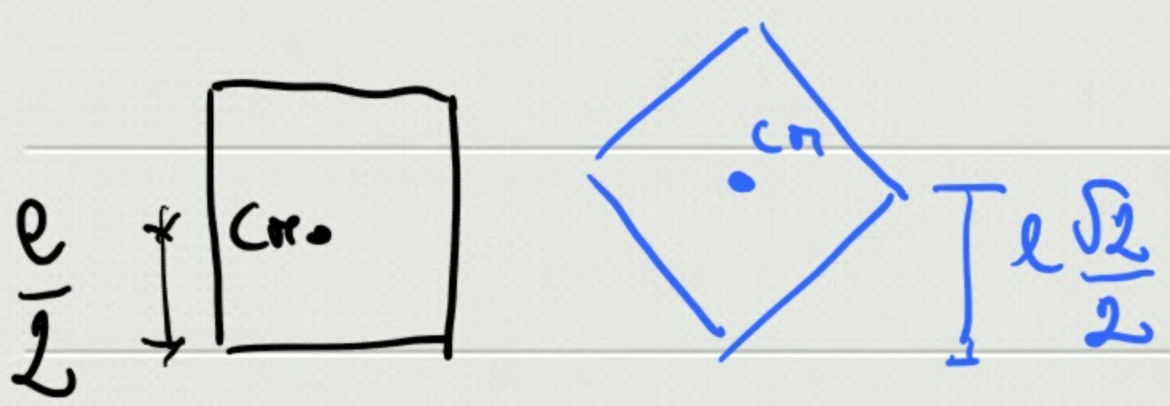
$$\overset{!}{=} 2{,}4 \cdot 10^4 \text{ J}$$

$\omega(\vartheta = 45°) = ?$

$$W_{TOT} = \Delta E_K \overset{E_{K_i}=0}{=} \frac{1}{2} I_z \omega^2$$

$$= W_F + W_{peso}$$

$$\hookrightarrow = -\Delta E_{p,peso}$$



$$-\Delta E_p = -\left(mg\frac{l\sqrt{2}}{2} - mg\frac{l}{2}\right) = -mgl\left(\frac{\sqrt{2}}{2} - 1\right)$$

$$\Rightarrow Fl + mg\frac{l}{2}\left(1-\sqrt{2}\right) = \frac{1}{2} I_z \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{3\left[2F - mg\left(\sqrt{2}-1\right)\right]}{2ml}} = 2{,}72 \text{ rad/s}$$